

Campbell Downs

West Lafayette, IN | Brentwood, TN

📞 615-762-6119

✉ campbellndowns@gmail.com

🌐 [linkedin.com/in/campbell-downs](https://www.linkedin.com/in/campbell-downs)

Education

Purdue University

Expected May 2029

Bachelor of Science in Aerospace Engineering; Minor in Product Lifecycle Management

West Lafayette, IN

- **GPA: 3.31** | U.S. Citizen
- **Relevant Coursework:** Aerospace Design, Aeromechanics, Digital Manufacturing, Simulation & Visualization

Technical Skills

CAD / PLM: SolidWorks, Siemens NX, Teamcenter (PDM), parametric modeling, assemblies, ECO workflows
Analysis: SolidWorks Simulation, linear static FEA, fatigue analysis, transient thermal analysis, design optimization
Programming / Data: MATLAB, Python, C, C#, JavaScript, HTML, Microsoft Excel
Manufacturing: 3D printing, additive manufacturing, introductory CNC machining, general shop tools

Certifications

Certified SOLIDWORKS Associate (CSWA) – Dassault Systèmes, September 2026

Experience & Engineering Involvement

Handshake AI

May 2026 – Present

AI Fellow (Independent Contractor)

Remote

- Completed **350+** **contracted hours** evaluating AI outputs and annotating video data for multimodal action-recognition and tracking models.
- Averaged approximately **4/5** on quality assessments, advanced twice to higher work-access tiers, and performed peer quality-control reviews.

Purdue Orbital

September 2026 – Present

Launch Structures Team Member

West Lafayette, IN

- Supporting the **HAPSIS II** landing-structure redesign for approximately **6 ft/s impact velocity**, balancing impact survivability with reduced structural mass and helium requirements.

Nashville Adventure Science Center

2025 – 2026

STEM Education Volunteer

Nashville, TN

- Contributed **40+** **volunteer hours** supporting STEM education, science outreach, and visitor engagement.

Selected Engineering Projects | Portfolio

Project ECHO – Agricultural Mapping UAV | *Python, CAD, Systems Optimization*

2026 – Present

- Developing a sub-**\$5,000** agricultural mapping UAS targeting **35+ minimum endurance** and **≤2 cm/px GSD at 70 m AGL**.
- Performing propulsion, imaging, and subsystem trade studies to balance endurance, system mass, image quality, and total platform cost.

Multi-Station Fitness Rack | *SolidWorks, Structural FEA, Fatigue Analysis*

2026

- Designed an A36 steel fitness rack for a **250 lb user load** and validated four structural load cases using static FEA and fatigue analysis.
- Redesigned critical members based on simulation results until the assembly satisfied defined structural and fatigue requirements.

Bicycle Horn Mount Optimization | *SolidWorks, Parametric & Topology Optimization*

2026

- Evaluated **270 parametric configurations** of a Nylon 6/10 mount across five variables for mass, stiffness, and factor of safety.
- Combined parametric and topology optimization to reduce mass by approximately **30%** while maintaining FOS above 100.

Interests

Golf, strength training, baseball, and hiking